

Intelligent Complaint Classification Using Transformer Architectures in Finance Sector

Background and Motivation

Customer satisfaction has become one of the most important determinants of organizational success in today's highly competitive business environment. Whether in banking, telecommunications, healthcare, e-commerce, government services, or transportation, organizations receive thousands of customer complaints every day through emails, websites, social media, and support centers. Efficient complaint management directly influences customer trust, brand reputation, operational efficiency, and long-term business growth.

Unfortunately, many organizations still rely on manual complaint triage or simple keyword-based routing systems. These approaches are often slow, inconsistent, and unable to capture the true meaning or urgency of customer concerns. Customers frequently describe the same issue using different vocabulary, informal language, abbreviations, or emotional expressions, making rule-based systems unreliable. As complaint volumes continue to increase, manual processing becomes increasingly expensive and delays the delivery of appropriate support, ultimately reducing customer satisfaction.

Recent developments in transformer-based language models, such as BERT, have significantly improved the ability of machines to understand contextual relationships and semantic meaning in natural language. Unlike traditional approaches that depend on isolated keywords, transformer architectures analyze the entire context of a sentence, enabling them to identify the true intent behind customer complaints even when they are expressed indirectly or contain spelling mistakes and colloquial language.

This project proposes the development of an intelligent complaint classification system using a fine-tuned transformer model capable of automatically categorizing customer complaints into appropriate service departments. By preprocessing real-world complaint datasets, addressing challenges such as noisy text and class imbalance, and leveraging the contextual understanding of transformer architectures, the proposed system aims to significantly improve complaint routing accuracy while reducing manual workload.

The potential impact of this project extends beyond technical performance. Automated complaint classification can substantially improve customer experience by reducing response times and ensuring that issues are directed to the appropriate teams more quickly. Faster resolution enhances customer loyalty, strengthens organizational reputation, and enables businesses to allocate human resources more efficiently. Small and medium-sized enterprises, which often lack large customer service departments, could particularly benefit from affordable AI-powered complaint management systems.

PROJECT SCOPING DOCUMENT

Automated Complaint Classification in Financial industry with Transformer Models

Project Title	Intelligent Complaint Classification Using Transformer Architectures in Finance Sector
Sector Focus	Banking / Financial Services - Consumer complaint triage
Project Scope	Single language (English), compare two transformer models
Document Purpose	Define project scope, objectives, and the key questions the team must answer before development begins
Date	25th July 2026

1. Background and Problem Statement

Banking institutions receive a high volume of unstructured customer complaints daily, across channels such as call centre transcripts, emails, mobile app feedback, and branch reports. Manual triage of this volume is slow, inconsistent, and expensive, and traditional keyword-based routing struggles to capture nuance, sentiment, and the true underlying intent of a complaint (such as distinguishing a billing dispute from a fraud allegation that happens to mention a similar term).

This project will design, fine-tune, and evaluate a transformer-based text classification pipeline that automatically categorizes banking complaints into actionable classes, reducing manual triage time and improving resolution efficiency.

2. Project Scope

2.1 In Scope

1. Gathering and refining a publicly accessible corpus of consumer grievances sourced from banking entities.
2. Sanitizing and organizing input data by mitigating textual noise, encompassing typographical errors, shorthand, colloquialisms, polyglot nuances, and skewed category distributions.

3. Engineering and optimizing a pre-trained two transformer architecture (DistilBERT and RoBERT) to facilitate the automated sorting of banking grievances into established operational silos.
4. Constructing a domain-specific hierarchy including account administration, unauthorized activity, credit products, transactional discrepancies, card management, digital platforms, and general service quality.
5. Validating the predictive framework through rigorous benchmarking of precision, recall, F1-scores, and detailed error distribution via confusion matrices.

2.2 Out of Scope

- Full production deployment into a live banking IT system.
- Automated complaint resolution or response generation (this project classifies, it does not resolve).
- Real-time streaming ingestion from live banking channels.
- Future integration of multilingual support beyond the language(s) explicitly chosen by the team.

3. Objectives

3.1 General Objective

To design and deliver an end-to-end pipeline that automatically classifies banking-sector consumer complaints using a baseline and one main model fine-tuned transformer model, demonstrably reducing manual triage effort while maintaining high classification accuracy.

3.2 Specific Objectives

- Source for a well-labelled dataset of banking complaints with a clearly defined category taxonomy.
- Preprocess the text to handle real-world noise (spelling errors, informal language, incomplete sentences).
- Fine-tune a pre-trained transformer model suited to the resource constraints of the project (compute, data volume, language).

- Address class imbalance so that minority but high-severity categories (e.g. fraud) are not under-detected.
- Evaluate the model rigorously across precision, recall, and F1-score per class, not just overall accuracy.
- Produce a short report or dashboard demonstrating the model's practical impact on triage time and routing accuracy.

4.1 Project Context

1. We aim to train the model with available [US English dataset](#) and test on the freely available African English customer complaints.

1. The Bank Customer Complaint Analysis dataset will be used as the primary labeled training corpus. An independent corpus of African English banking complaints collected from publicly available sources will be used exclusively as an external test set to evaluate cross-domain generalization
2. The targeted users are the Customer Service Representatives, Complaint Resolution Teams, bank managers, Customer Experience (CX) Teams
3. The decision made manually on the complaints are reading each complaint, identifying its category, determining its priority and assigning it to the appropriate department
4. Customers complaints are usually received via email, mobile banking app, social media, call center and web forms
5. Tools adopted lately are CRM systems, excel, ticketing system and internal banking complaint management software
6. Possible issues with dealing with customers complaints in the banking systems are complaints routed to the wrong department, delayed response due to high complaint volumes, human errors, difficulty understanding multilingual or informal complaints, inconsistent categorization between staff members.
7. Success in this project can be measured in terms of operational efficiency, customer satisfaction, and cost savings.

4.2 Complaint Taxonomy & Labeling

1. Complaint categories will be based on existing banking complaint taxonomies and adapted to the African banking context.

2. The model will perform single-label classification, assigning each complaint to one main category.
3. Severity or urgency prediction is outside the project scope and will be considered as a separate task.
4. The project team will handle labeling using predefined guidelines and quality checks for consistency.
5. The number of categories will depend on dataset size and sufficient samples per class.

4.3 Data Sourcing & Preprocessing

1. The project will use a publicly available banking complaints dataset.
2. Complaints will be in English, while it would be tested on Africa English banking complaints.
3. The dataset size will be evaluated to ensure sufficient samples for transformer fine-tuning.
4. Class imbalance will be handled using techniques such as class weighting, oversampling, or data augmentation.
5. Text preprocessing will include cleaning, normalization, and removal of irrelevant or sensitive information.
6. Any personally identifiable information will be anonymized before model training and evaluation.

4.4 Model Selection & Fine-Tuning

1. This project will adopt the DistilBERT , and RoBERTa pretrained model, DistilBERT is efficient and light weight baseline and RoBERTa is a strong performer in English.
2. This project will be developed using a local machine and Google Colab for experimentation and model training. Mostly, GPU access will be used to accelerate fine-tuning of pretrained transformer models.
3. Training will be constrained by batch size, sequence length, and the number of epochs to keep runtime manageable. If GPU access is limited, model training may be done on a reduced dataset or with parameter-efficient fine-tuning (LoRA) to reduce training time and memory usage.

4. Customer complaints submitted through mobile banking applications, email, web forms, and social media are typically concise, usually ranging from **20 to 150 words**. Therefore, a **maximum input sequence length of 256 tokens** is expected to be sufficient for the majority of complaints while balancing computational efficiency and memory usage.
5. During preprocessing, tokenization will be performed using the tokenizer associated with the selected pretrained transformer model (RoBERTa). Complaints longer than the maximum sequence length will be truncated, preserving the beginning of the complaint where the main issue is usually described. Since banking complaints are generally short, the proportion of truncated samples is expected to be minimal. Consequently, chunking will not be implemented, as it would introduce unnecessary complexity without significant performance benefits.
6. To evaluate the effectiveness of the proposed transformer-based approach, this project will compare the performance of two pretrained transformer models: DistilBERT and RoBERTa. DistilBERT will serve as the baseline transformer model due to its lightweight architecture and computational efficiency, while RoBERTa will serve as the primary model because of its demonstrated strong performance on English-language natural language processing tasks.
7. Both models will be fine-tuned on the bank customer complaint dataset, and their performance will be evaluated and compared using standard text classification metrics, including accuracy, precision, recall, and F1-score.

4.5 Evaluation Framework

1. The dataset will be divided into training, validation, and testing sets using a stratified split to maintain the original distribution of complaint categories. This ensures that each class is properly represented during training and evaluation.
2. Model performance will be evaluated using multiple metrics, including accuracy, macro F1-score, weighted F1-score, and recall, especially for critical complaint categories such as fraud-related cases. Macro F1 will be prioritized because it provides a balanced evaluation across all classes, including minority categories.
3. The model will be considered acceptable if it achieves an overall F1-score of at least **85%** on the test dataset while maintaining balanced precision and recall across the complaint categories. In addition, RoBERTa should demonstrate performance that is

comparable to or better than DistilBERT, thereby indicating its effectiveness for bank customer complaint classification.

4. Misclassifications will be analyzed through confusion matrices, manual review of incorrectly classified complaints, and identification of common category confusion patterns. These analyses will help identify weaknesses in the model and guide further improvements.
5. The reduction in manual triage time will be estimated by comparing the time required for manual complaint categorization with the time taken when using the automated classification system. The evaluation will consider improvements in processing speed, routing accuracy, and reduction of repetitive manual tasks.

4.6 Deployment, Integration & Presentation

A simple working demo will be developed as a web interface where users can enter a complaint and receive an automatic classification.

The output will include the predicted complaint category, the confidence score will be calibrated with Temperature Scaling to validate the ideal output.

The final deliverables will include a research report, the source code, a Jupyter notebook for model training and evaluation, and a simple web demo.

4.7 Ethics, Privacy & Risk

1. In banking, a wrong label isn't just an "error." Imagine a fraud complaint getting tagged as a "minor service issue." That customer could lose money while waiting, and the bank loses trust.

To account for that, we'll design the model with 3 safeguards.

Urgency flags: High -risk categories like fraud, scams, and account takeovers get priority routing

Human-in-the-loop: Borderline or high- impact cases always get reviewed by a person before closing.

Confidence thresholds: Raw model scores are overconfident, so we first calibrate them using a validation set. Any complaint below the set confidence threshold is escalated to a human reviewer instead of auto- routing, to prevent misrouting urgent or high-risk cases.

2. Customer privacy will be protected by anonymizing personal information, removing sensitive data, and ensuring secure data storage and handling throughout the model development process.

3. Any usage restrictions associated with public datasets will be reviewed and followed, ensuring compliance with licensing requirements and responsible data usage practices.

5. 1 Expected Deliverables

- Cleaned, labelled banking complaint dataset with documented taxonomy.
- Preprocessing pipeline (code + documentation of design choices).
- Fine-tuned transformer model with saved weights/checkpoint.
- Evaluation report: precision, recall, F1-score per class, confusion matrix, and comparison against a baseline model.
- Short demo showing the model classifying a new complaint.
- Final written report summarizing scope, methodology, results, and limitations.

5.2. Assumptions & Open Risks

- Dataset availability and size may constrain how many complaint categories can realistically be supported especially when checked for its performance on Africa dataset (English)
- Compute resource limits may require using a smaller/distilled model rather than a full-size transformer.

5.3. Timeline

Week 1: Scoping & Dataset Setup (Challenges 1 & 2)

- Finalize Problem Statement and source complaint dataset.
- Clean text, filter categories (>500 samples), and complete the Data Card .
- **Deliverable:** Approved problem scope, clean dataset, and Data Card.

Week 2: Impact Strategy & Preprocessing

- Draft the Impact Statement Card and Mini-Engagement Plan .
- Build text preprocessing, tokenization (256 length), and class-balancing pipelines.
- **Deliverable:** Completed Challenges 3 & 4 and ready-to-train data splits.

Week 3: Transformer Model Fine-Tuning

- Set up Colab GPU environment.
- Fine-tune DistilBERT (baseline) and RoBERTa-base (primary model).
- **Deliverable:** Saved model checkpoints and training logs.

Week 4: Evaluation, Calibration & Safety Safeguards

- Benchmark models using Macro F1, Recall, and confusion matrices.
- Calibrate confidence scores (Temperature Scaling) and set human-in-the-loop thresholds.
- **Deliverable:** Model evaluation report and safety escalation logic.

Week 5: Prototype Deployment & Final Report

- Build a simple web demo displaying predictions and confidence scores.
- Package repository code, model weights, and Jupyter notebooks.
- **Deliverable:** Live web prototype and final research report.

5.4. Proposed Dataset (Links)

<https://www.consumerfinance.gov/data-research/consumer-complaints/>

[Bank Customer Complaint Analysis](#)

[Automatic Ticket Classification](#) (78,313 customer complaints with 22 features)

To ensure reliable model training and evaluation, only complaint categories with at least **500 samples** will be included in this project. Categories with fewer samples will be excluded or merged where appropriate